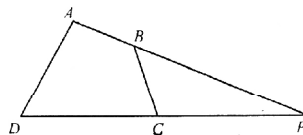


AMTI (NMTC) - 2010
GAUSS CONTEST - PRIMARY LEVEL
 (Standard - 5/6)

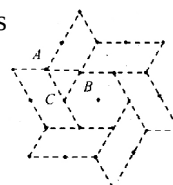
PART - A

Note :

- ◆ Only one of the choices A, B, C, D is correct for each question. Shade that alphabet of your choice in the response sheet. (If you have any doubt in the method of answering, seek the guidance of your supervisor).
 - ◆ For each correct response you get 1 mark; for each incorrect response you lose $\frac{1}{4}$ mark.
1. n ; a are natural numbers each greater than 1. If $a + a + \dots + a = 2010$, and there are n terms on the left hand side, then the number of ordered pairs (a, n) is
 (A) 7 (B) 8 (C) 14 (D) 16
 2. X is a seven digit number. Y is an eight digit number 5 more than X . The number of possible values of Y is
 (A) 5 (B) 4 (C) 1 (D) 3
 3. The sum of the digits of a four digit number is 3. The difference between the biggest and the smallest of these numbers is
 (A) 1998 (B) 1989 (C) 1899 (D) 1809
 4. ABCD is a quadrilateral $AB = AD$, $BC = CD$. $\angle BAD = \angle BDC = 20^\circ$. The measure of the angles $\angle ABC$, $\angle BCD$ and $\angle CDA$ are respectively.
 (A) 100° , 140° , 100° (B) 20° , 140° , 100°
 (C) 100° , 100° , 20° (D) 140° , 100° , 100°
 5. The digital sum of a certain number is 2010. The minimum possible number of digits is
 (A) 223 (B) 224 (C) 2009 (D) 2010
 6. In the diagram ABCD is a quadrilateral. $\angle ABC = 150^\circ$, $\angle DAB = \frac{1}{3} \angle ABC$ and $\angle BCD = 60^\circ$. Then $\angle ADP$ and $\angle APD$ are respectively



- (A) 100° and 30° (B) 110° and 20° (C) 80° and 40° (D) 120° and 10°
7. Given two addition problems
 $a = 1 + 12 + 123 + \dots + 123456789$
 $b = 987654321 + 87654321 + \dots + 21 + 1$
 The digits in the hundredth place of a and b are respectively
 (A) 4 and 6 (B) 1 and 6 (C) 4 and 4 (D) 1 and 4
8. The number of numbers with 2010 digits is
 (A) $\underbrace{999\dots90}_{2009 \text{ times}}$ (B) $\underbrace{999\dots99}_{2010 \text{ times}}$ (C) $\underbrace{900\dots00}_{2009 \text{ times}}$ (D) $\underbrace{900\dots00}_{2010 \text{ times}}$
9. In the adjoining rangoli design the distance between any two adjacent dots is 1 unit. In the diagram we find the triangle ABC is equilateral. The number of smallest equilateral triangles thus formed by joining the dots suitably is



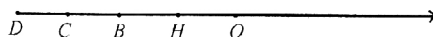
- (A) 24 (B) 28 (C) 15 (D) 30

10. The natural numbers are written in the following form

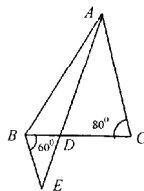
First row	1
Second row	2 3 4
Third row	9 8 7 6 5
Fourth row	10 11 12 13 14 15 16
.....	

The number 2010 is in the

- (A) 45th row as the 72nd number from the right.
 (B) 44th row as the 73rd number from the right.
 (C) 45th row as the 16th number from the left.
 (D) 44th row as the 17th number from the left.
11. The number of ways of labeling the ray using the points shown in the figure is



- (A) 6 (B) 4 (C) 1 (D) 10
12. ABC is a triangle in which $\angle BAC = 60^\circ$, $\angle ACB = 80^\circ$, ADE is the angle bisector of $\angle BAC$. Then the triangle BDE is



- (A) Isosceles (B) Equilateral (C) Right angled (D) Scalene
13. The digits of a three digit number are 3, 7 and x in that order and $37x = 3^3 + 7^3 + x^3$. The value of x is
 (A) 1 or 2 (B) 0 or 2 (C) 1 or 2 (D) 0; 1 or 2
14. In the sequence 1, 4, 3, 6, 5, 8, 7, 10, ... we have $t_{2n-1} = 2n-1$ and $t_{2n} = t_{2n-1} + 3$ [(ie) every odd term is that odd number and the next even term is 3 more than the previous odd term]. If $t_m = 2010$, then m is equal to
 (A) 1005 (B) 1004 (C) 2008 (D) 2010
15. The largest of the four numbers given below is
 (A) $3.\overline{1416}$ (B) $3.1\overline{416}$ (C) $3.14\overline{16}$ (D) 3.1416

PART - B

Note :

- ♦ Write the correct answer in the space provided in the response sheet.
 - ♦ For each correct response you get 1 mark; for each incorrect response you lose $\frac{1}{2}$ mark.
16. The percentage of the square numbers among the numbers between 9 and 100 is _____.
17. In a sequence, the first term $t_1 = 6$, $t_2 = a + 3$, $t_3 = 42$, $t_{n+3} = 3t_{n+2} - 2t_{n+1}$ for $n = 1, 2, \dots$; then every term of the sequence is a multiple of _____.
18. a234 is a four digit number which is divisible by 18 then a is _____.
19. a, b, c are squares of three consecutive integers and $(b-a) = 87$ then c is _____.
20. In the sum of $1 + 11 + 111 + \dots + 111111111$ the digit that does not occur is _____.
21. After simplification the denominator of the fraction $\frac{57}{10} \times \frac{58}{9} \times \frac{59}{8} \times \frac{60}{7} \times \frac{61}{6} \times \frac{62}{5} \times \frac{63}{4} \times \frac{64}{3}$ is _____.
22. In the adjoining figure the value of $x + y - z$ is _____.
23. The number of ways in which 100 can be written as the sum of two prime numbers is _____.
24. The number of natural numbers (a, b) satisfying the relation $7 + a + b = 10$ is _____.
25. A boy divided a certain number by 75 instead of by 72 and got both quotient and remainder to be 72. What should be the quotient and remainder if it is divided by 72 _____.

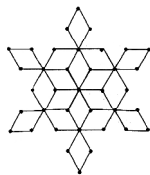
AMTI (NMTC) - 2010
GAUSS CONTEST - PRIMARY LEVEL
(Standard - 5/6)

PART - A

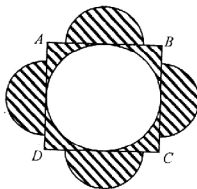
Note :

- ♦ Only one of the choices A, B, C, D is correct for each question. Shade that alphabet of your choice in the response sheet. (If you have any doubt in the method of answering, seek the guidance of your supervisor).
- ♦ For each correct response you get 1 mark; for each incorrect response you lose $\frac{1}{4}$ mark.

1. Given a sequence of two digit numbers grouped in brackets as follows :
(10), (11, 20), (12, 21, 30), (13, 23, 31, 40) ... (89, 98), (99).
The digital sum of the numbers in the bracket having maximum numbers is
(A) 9 (B) 10 (C) 9 or 10 (D) 18
2. Using the digits 2 and 7, and addition or subtraction operations only, the number 2010 is written. The maximum number of 7 that can be used, so that the total numbers used is a minimum is
(A) 284 (B) 286 (C) 288 (D) 290
3. In the adjoining rangoli design each of the four sided figures is a rhombus and the distance between any two dots is 1 unit. The total area of the design is



- (A) $36\sqrt{3}$ (B) $9\sqrt{3}$ (C) $24\sqrt{3}$ (D) $18\sqrt{3}$
4. In the addition problem shown, different letters represent different digits. If the carry over from adding the units digit is 2, then $(A + I)$ cannot be
(A) 2 (B) 4 (C) 7 (D) 9
 5. The percentage of natural numbers form 10 to 99 both inclusive which are the product of consecutive natural numbers is
(A) $9\frac{7}{9}$ (B) $7\frac{7}{9}$ (C) 10 (D) 9
 6. In the adjoining figure, ABCD is a square of side 4 units. Semicircles are drawn outside the squares with diameter 2 units as shown. The area of the shaded portion in square units is



- (A) 8 (B) 16 (C) $16 - 2\pi$ (D) $8 - \pi$
7. $n = 1 + 11 + 111 + \dots + 1111111111$. The digital sum of n is
(A) 39 (B) 38 (C) 37 (D) 36
 8. If $A + B = C$, $B + C = D$, $D + A = E$ then $A + B + C$ is
(A) E (B) $D + E$ (C) $E - D$ (D) $B - D + C$
 9. A three digit number $ab7 = a^3 + b^3 + 7^3$. Then a is
(A) 6 (B) 4 (C) 7 (D) 8
 10. Among the participants of this screening test, some of them together got the correct answers for all the problems, not all of them got more than 8 problems correct. The maximum and minimum number of problems solved by the three together are
(A) 25, 25 (B) 24, 26 (C) 25, 27 (D) 25, 28

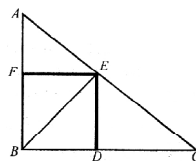
11. $\frac{1}{a} + \frac{1}{b} = \frac{1}{13}$ where a, b are natural numbers.
 (1) $a = b = 26$ (2) $a = 13, b = 13 \times 14$ (3) $a = 14, b = 13 \times 14$.
 Of these statements the correct statements are
 (A) (1) and (2) (B) (1) and (3) (C) (2) and (3) (D) (1) (2) and (3)
12. $a^3 + b^3 = p_1 \times p_2$, $a^3 - b^3 = p_3$, where p_1, p_2, p_3 are prime numbers, $a > b$, $2 \leq a, b < 5$.
 The following statements are also given :
 (1) $p_1 + p_2 + p_3$ is a prime number. (2) $p_3 - (p_1 + p_2)$ is a prime number. (3) $p_1 \sim p_2$ is a prime number. Then the values of a and b are respectively.
 (A) (3, 4) (B) (4, 3) (C) (3, 2) (D) (2, 3)
13. p is a prime number greater than 3. When p^2 is divided by 12 the remainder is
 (A) always an odd number greater than 2.
 (B) always 1.
 (C) 1 or 11.
 (D) always an even number.
14. The regular polygons have the number of sides in the ratio 3 : 2 and the interior angles in the ratio 10 : 9 in that order. The number of sides of the polygons are respectively
 (A) 6 and 4 (B) 9 and 6 (C) 12 and 8 (D) 15 and 10
15. a is a real number such that $a^3 + 4a - 8 = 0$. Then the value of $a^7 + 64a^2$ is
 (A) 128 (B) 164 (C) 256 (D) 180

PART - B

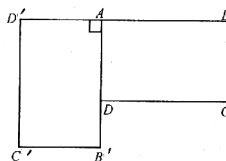
Note :

- ♦ Write the correct answer in the space provided in the response sheet.
- ♦ For each correct response you get 1 mark; for each incorrect response you lose $\frac{1}{2}$ mark.

16. ABC is a right angled triangle with $B = 90^\circ$. BDEF is a square. BE is perpendicular to AC. The measure of $\angle DEC$ is _____.



17. p is prime number and $p = a^2 - 1$. The number of divisors of $a + p$ is _____.
18. $a = 1 + 3 + 5 + 7 + \dots + 2009$
 $b = 2 + 4 + 6 + 8 + \dots + 2010$ then the value of $(a - b)^2$ is _____.
19. $1\frac{1}{2} + 2\frac{2}{3} + 3\frac{3}{4} + \dots + n\frac{n}{n+1}$, on complete simplification has the denominator _____.
20. The biggest value of $\frac{10a}{10+a}$ ($a \in N$) is never greater than _____.
21. From a point within an equilateral triangle perpendiculars are drawn to the three sides and are 5, 7 and 9 cms in length. The perimeter of the triangle is _____ cm.
22. The number of terms in the expansion $(a + b + c)^3$ is _____.
23. The value of x satisfying the equation $\frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{x}}}} = \frac{3}{4}$ is _____.
24. ABCD is a rectangle rotated clockwise about A by 90° as shown. The rotation takes B to B', C to C', D to D'. $AB = 6$ cm, $BC' = 10$ cm. The breadth of the rectangle ABCD is _____.



25. AB is a line segment 2000 cm long. The following design of semicircles is drawn on AB, with $AP = 5$ cm and repeating the designs. The area enclosed by the semicircular designs from A to B is _____.

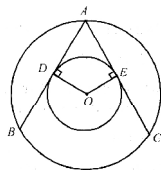
AMTI (NMTC) - 2010
GAUSS CONTEST - PRIMARY LEVEL
 (Standard - 5/6)

PART - A

Note :

- ♦ Only one of the choices A, B, C, D is correct for each question. Shade that alphabet of your choice in the response sheet. (If you have any doubt in the method of answering, seek the guidance of your supervisor).
- ♦ For each correct response you get 1 mark; for each incorrect response you lose $\frac{1}{4}$ mark.

1. The number which, when subtracted from the terms of ratio $a : b$ makes it equal to $c : d$ is
 (A) $\frac{ab - cd}{ab + cd}$ (B) $\frac{bc - ad}{c + d}$ (C) $\frac{ab + cd}{c + d}$ (D) $\frac{ab - cd}{b - c}$
2. In a Kilometer race Ram beats Shyam by 25 meters or 5 seconds. The time taken by Ram to complete the race is
 (A) 1 minute (B) 5 minutes and 30 seconds
 (C) 3 minutes and 15 seconds (D) 4 minutes and 10 seconds
3. Through D, the mid-point of the side BC of a triangle ABC, a straight line is drawn to meet AC at E and AB produced at F so that $AE = AF$. Then the ratio $BF : CE$ is
 (A) 1 : 2 (B) 2 : 1 (C) 1 : 3 (D) None of these
4. In the bigger of two concentric circles two chords AB and AC are drawn to touch the smaller circle at D at E. Then BC is equal to



- (A) 3DE (B) 4DE (C) 2DE (D) $\frac{3}{2}$ DE
5. The number of solutions of the equation $x^{\log_{10} x} = 100x$ is
 (A) 0 (B) 1 (C) 2 (D) 3
6. The internal bisector AE of the angle A of triangle ABC is
 (A) not greater than the median through A for all triangles.
 (B) not greater than the median through A for only acute angled triangles.
 (C) Not greater than the median through A for only obtuse angled triangles.
 (D) not less than the median through A for all triangles.
7. In the adjoining diagram ABC is an equilateral triangle and BCDE is a square. The side of the equilateral triangle is 2010. The radius of the circle is
 (A) 2010 (B) 4020 (C) 6030 (D) 8040
8. Given a and b are integers the expression $(a^2 + a + 2011)(2b + 1)$ is
 (A) Odd for exactly 2010 values of a.b.
 (B) Odd for all values of a, b.
 (C) Even for exactly one value of a and two values of b.
 (D) Odd for exactly for one value of a and one value of b.
9. A sequence of real numbers x_n is defined recursively as follows. x_0, x_1 are arbitrary positive real numbers and $x_{n+2} = \frac{1+x_{n+1}}{x_n}$ $n = 0, 1, 2, \dots$. Then the value of x_{2011} is
 (A) 1 (B) x_0 (C) x_1 (D) x_2
10. If $xy = 6$ and $x^2y + y^2x + x + y = 63$, the value of $x^2 + y^2$ is
 (A) 81 (B) 18 (C) 2010 (D) 78

11. If p is the perpendicular drawn from the vertex of a regular tetrahedron to the opposite face and if each edge is equal to 2 units, then p is
 (A) $8\sqrt{3}$ (B) $\frac{8\sqrt{3}}{2}$ (C) $\frac{8\sqrt{3}}{5}$ (D) $\frac{8\sqrt{3}}{3}$
12. The remainder when the polynomial $x + x^3 + x^9 + x^{27} + x^{81} + x^{243}$ is divided by $x^2 - 1$.
 (A) $6x$ (B) $2x$ (C) $3x$ (D) 1
13. Consider the sequence 4, 4, 8, 2, 0, 2, 2, 4, 6, 0, where the n^{th} term is the units place of the sum of the previous two terms for $n \geq 3$. If S_n is the sum to n terms of this sequence, then the smallest 'n' for which $S_n > 2010$ is
 (A) 253 (B) 502 (C) 503 (D) 504
14. P is a point inside an equilateral triangle of side 2010 units. The sum of the lengths of the perpendiculars drawn from P to the sides is equal to
 (A) 2010 (B) $2010\sqrt{3}$ (C) $1005\sqrt{3}$ (D) $\frac{2010}{\sqrt{3}}$
15. The equation $\log_{2x}\left(\frac{2}{x}\right)(\log_2 x)^2 + (\log_2 x)^4 = 1$ has
 (A) A root less than 1. (B) Has only one root greater than 1
 (C) Two irrational roots. (D) No real roots.

PART - B

Note :

- ◆ Write the correct answer in the space provided in the response sheet.
 - ◆ For each correct response you get 1 mark; for each incorrect response you lose $\frac{1}{2}$ mark.
16. The value of $\sqrt[3]{20+14\sqrt{2}} + \sqrt[3]{20-14\sqrt{2}}$ is _____.
17. If a, b are positive and $a + b = 1$ the minimum value of $a^4 + b^4$ is _____.
18. The whole surface area of rectangular block is 1332 cm^2 . The length, breadth and height are in the ratio 6 : 5 : 4. The sum of the length, breadth and height is _____ centimeters.
19. If $|x| + x + y = 10$, $x + |y| - y = 12$ then $x + y =$ _____.
20. Two parallel sides of a trapezoid are 3 and 9, the non parallel sides are 4 and 6. A line parallel to the bases (parallel sides) divides the trapezoid in to two trapezoids of equal perimeters. The ratio in which each of the non-parallel sides is divided is _____.
21. Triangle ABC has $AB = 17$, $AC = 25$ and the altitude to BC has length 15. The sum of the possible values of BC is _____.
22. $\frac{5}{6} = \frac{a_2}{2!} + \frac{a_3}{3!} + \frac{a_4}{4!} + \frac{a_5}{5!} + \frac{a_6}{6!}$, where $0 \leq a_i < i$, $i = 1, 2, 3, 4, 5, 6$. Then $a_2 + a_3 + a_4 + a_5 + a_6$ is _____.
23. A circle is circumscribed about a triangle with sides 30, 34, 16. It divides the circle into 4 regions with the non triangular regions being A, B, C; C being the largest. Then the value of $(C - A - B)$ is _____.
24. If a number n is divisible by 8 and 30, then the smallest number of divisors that n has is _____.
25. Both the roots of the quadratic equation $x^2 - 12x + K = 0$ are prime numbers. The sum of all such values of K is _____.
26. In a convex polygon of 16 sides the maximum number of angles which can all be equal to 10° is _____.
27. If an arc of circle 1 subtending 60° at the centre, has double the length as the arc subtending 75° at the centre in circle 2, then $\frac{\text{area of circle 1}}{\text{area of circle 2}}$ is _____.
28. A two digit number is equal to the sum of the product of its digits and the sum of its digits. Then the units place of the number is _____.
29. Let $f(x)$ be a polynomial of degree 1. If $f(10) - f(5) = 15$, then $f(20) - f(5)$ equals _____.
30. The number of perfect square divisors of the number $12!$ is _____.